

Data-Driven Dynamics, System Identification and Applications

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Overview for today's seminar

- Data-Driven Modal techniques review
 - **Proper Orthogonal Decomposition** (POD)
 - **Dynamic Mode Decomposition**
- POD/DMD for **Linear System Identification** (LSI)
- Towards non-linear systems: **Koopman** operator theory
- DMD and **Extended DMD** (EDMD) as algorithms for Koopman
- **Nonlinear Identification:**
 - **Sparse Identification of Nonlinear Dynamical systems** (SINDy)
- Perspectives for Offshore Mechanics

What is Data-Driven Dynamics/SI good for?

- To **extract** “most important” features/structures withing a dataset
 - **Proper Orthogonal Decomposition** (POD)
- To decompose the signal in **dynamically-interpretable** “modes”
 - **Dynamic Mode Decomposition**
- To **identify** the structure of a dynamical system entirely from data
 - **Sparse Identification of Nonlinear Dynamical systems** (SINDy)

Useful when no **robust, accurate** mathematical model is at hand

Once this is done, we can do: **Optimization, Control, Data-Assimilation/Forecast**

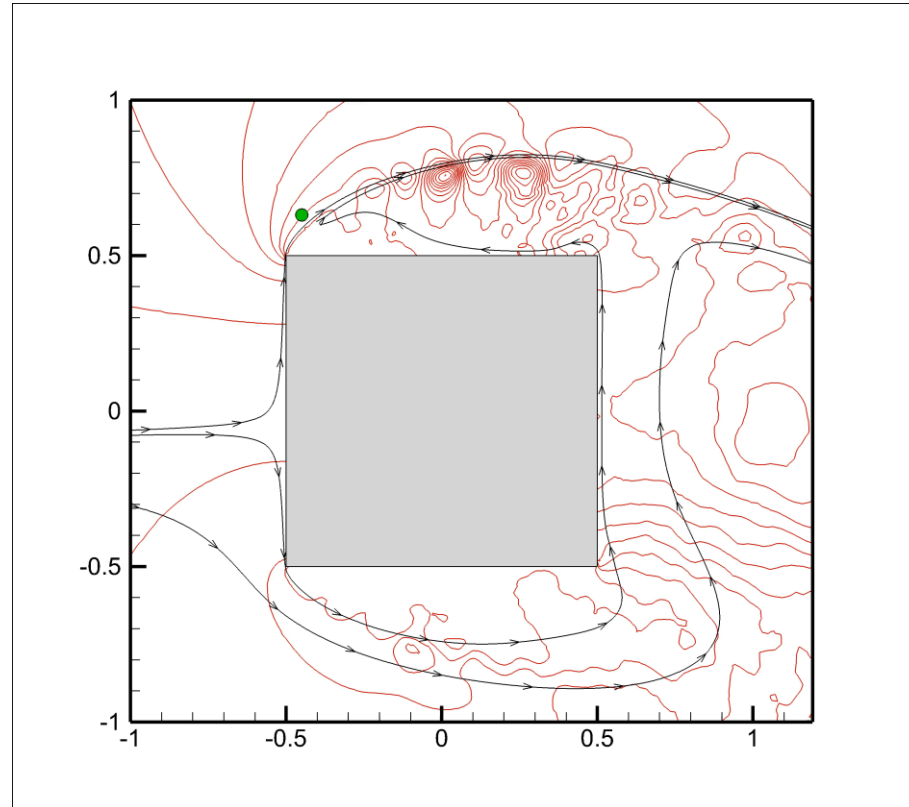
What is Data-Driven Dynamics/SI good for?

Oftentimes, those problems are **much harder** than it sounds!!!

Direct Numerical Simulation

$$Re = \frac{U_{\infty} L}{\nu} = 22000$$

Franceschini et al 2022



Proper Orthogonal Decomposition

Main Goal

To represent **highly dimensional** datasets (numerical or experimental) with as **fewest “structures”** as possible?

Obs.: This problem first arose, and was solved, in turbulence/fluid mechanics research (Lumley 1967, Sirovich 1987, Berkooz et al 1991, Towne et al 2019)

Proper Orthogonal Decomposition

Inputs (dataset): For every realization ξ , we have $\mathbf{q}(\xi) \in \mathbf{R}^N$

Output (mode): Deterministic (independent on ξ) modes $\boldsymbol{\phi} \in \mathbf{R}^N$

Those modes have to extremize the following:

$$\lambda = \frac{E[|\mathbf{q}(\xi)^{*,T} \boldsymbol{\phi}|^2]}{\boldsymbol{\phi}^{*,T} \boldsymbol{\phi}}$$

Where $E[\cdot]$ is an “expected-value” operator. For example:

$$E[\mathbf{q}] = \frac{1}{N_r} \sum_{\xi} \mathbf{q}(\xi)$$

Proper Orthogonal Decomposition

Then, this problem is equivalent to solving the following EVP:

$$\mathbf{C}\boldsymbol{\phi}_k = \lambda_k \boldsymbol{\phi}_k$$

With $\mathbf{C} = E[\mathbf{q}(\xi)\mathbf{q}^{*,T}(\xi)]$ the **two-point correlation tensor**

Numerically, we typically have vectors $\mathbf{q}(\xi)$, that are arranged in:

$$\mathbf{Q} = [\mathbf{q}(1), \dots, \mathbf{q}(N_r)]$$

The EVP becomes equivalent to find the SVD of this matrix:

$$\mathbf{Q} = \boldsymbol{\Phi}\boldsymbol{\Sigma}\mathbf{A}$$

Proper Orthogonal Decomposition

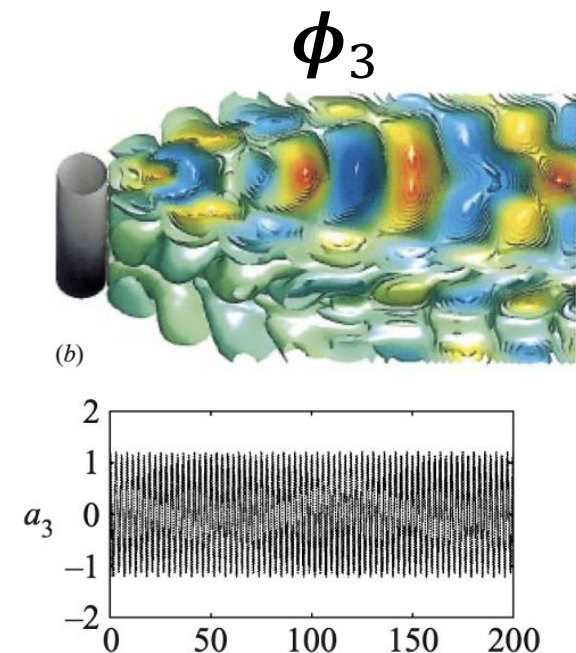
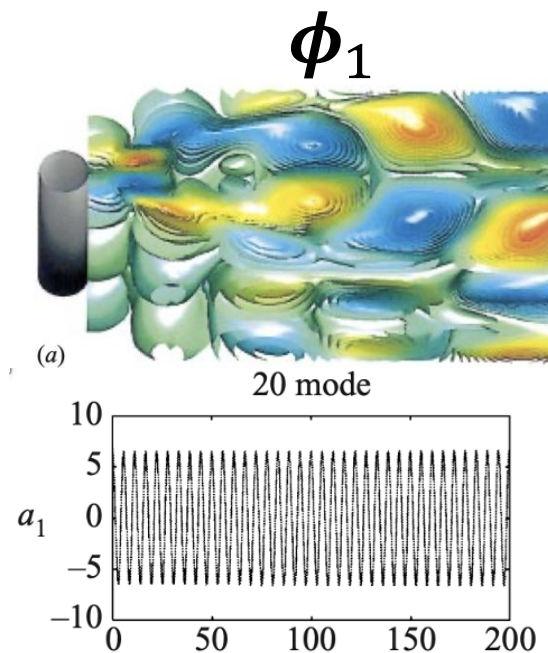
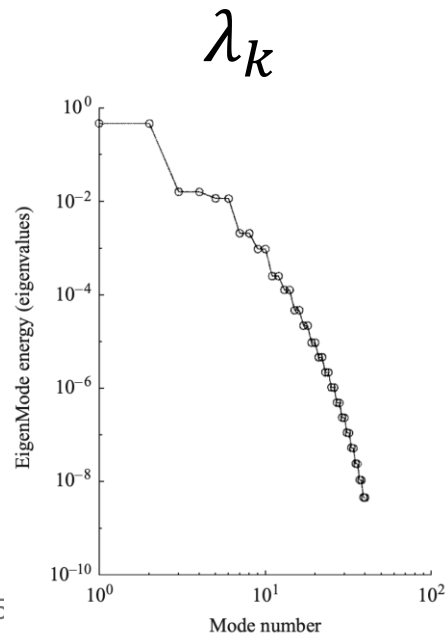
- Deterministic orthogonal modes $\boldsymbol{\phi}_m^{*,T} \boldsymbol{\phi}_n = \delta_{mn}$:
 $\boldsymbol{\phi} = [\boldsymbol{\phi}_1, \boldsymbol{\phi}_2, \dots]$
- Stochastic uncorrelated modes $E[a_m a_n] = \delta_{mn}$:
 $\mathbf{A} = [a_1(\xi), a_2(\xi), \dots]$
- Energies:
 $\boldsymbol{\Sigma} = \text{diag}(\lambda_1, \lambda_2, \dots)$

POD: Snapshot POD for flow around cylinder

If we consider each snapshot of a CFD/PIV dataset as a **realization**, this leads to the **snapshot-POD** or **space-only-POD**:

$$\mathbf{Q} = [\mathbf{q}(t_1), \dots, \mathbf{q}(t_{N_r})] = \boldsymbol{\phi}(\mathbf{x})\boldsymbol{\Sigma}\mathbf{A}(t)$$

Flow around cylinder, $Re = 185$ (Ma & Karniadakis 2002)



POD: Ex.: Eigenfaces

Turk Pentland 1991: Facial pattern determination for recognition
Principal Component Analysis (PCA), Karhuen-Loeve (KL)



POD/PCA/KL
Modes



Figure 1. (b) The average face Ψ .



Figure 2. Seven of the eigenfaces calculated from the input images of Figure 1.

Dynamic Mode Decomposition

Although (**snapshot**)-POD extracts energetic modes, they **do not** necessarily belong to the **same dynamics** $\{a_k(t)\}$

Indeed, permutations of $\mathbf{Q}$ lead to same $\phi(\mathbf{x})\Sigma$:

$$\begin{aligned}\mathbf{Q} &= [\mathbf{q}(t_1), \mathbf{q}(t_2), \dots, \mathbf{q}(t_{N_r})] = (\phi(\mathbf{x})\Sigma) \cdot \mathbf{A}(t) \\ \tilde{\mathbf{Q}} &= [\mathbf{q}(t_{147}), \mathbf{q}(t_{N_r}), \dots, \mathbf{q}(t_1)] = (\phi(\mathbf{x})\Sigma) \cdot \tilde{\mathbf{A}}(t)\end{aligned}$$

Dynamic Mode Decomposition

New Goal

Inspired (Schmid 2010) by **linear stability theory**, we wish to represent a signal $q(x, t)$ by **modes** $\hat{q}(x)$ and **time-varying dynamics** $e^{\lambda t}$

Obs.: This idea is rather mature in the field of **Linear System Identification (LSI)** field (to be discussed shortly)

The innovation from Schmid 2010 was the algorithm and its application to **nonlinear phenomena** (fluid mechanics)

Dynamic Mode Decomposition

To do so, we will give ourselves :

$$\mathbf{Q}^- = [\mathbf{q}(t_0), \dots, \mathbf{q}(t_{N-1})], \quad \mathbf{Q}^+ = [\mathbf{q}(t_1), \dots, \mathbf{q}(t_N)]$$

And we wish to establish a **linear dynamical relation** between those :

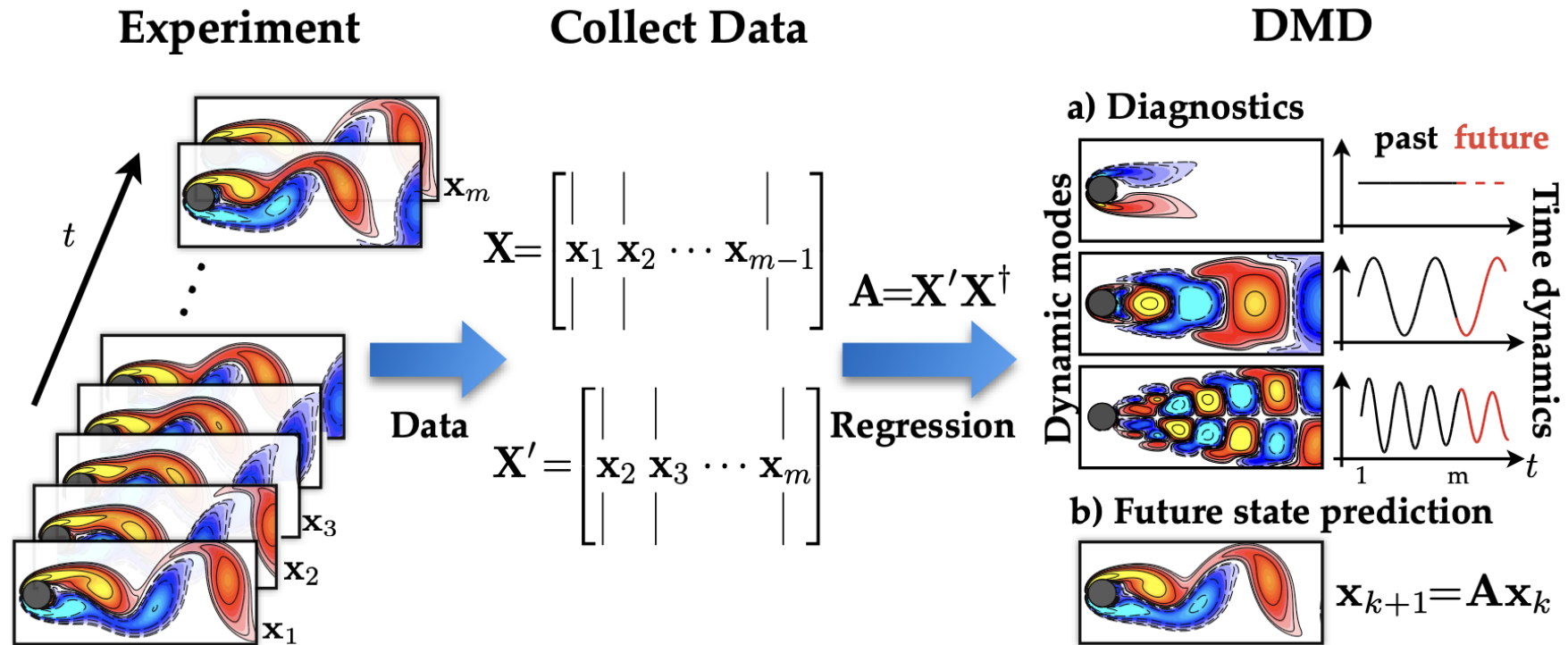
$$\mathbf{L}\mathbf{q}(t_k) = \mathbf{q}(t_{k+1}) \Rightarrow \mathbf{L}\mathbf{Q}^- = \mathbf{Q}^+$$

In LSI context, this is done by the Moore-Penrose pseudo-inverse :

$$\mathbf{L} = \mathbf{Q}^+(\mathbf{Q}^-)^\dagger = \mathbf{\Psi}\mathbf{\Lambda}\mathbf{\Psi}^{*,T}$$

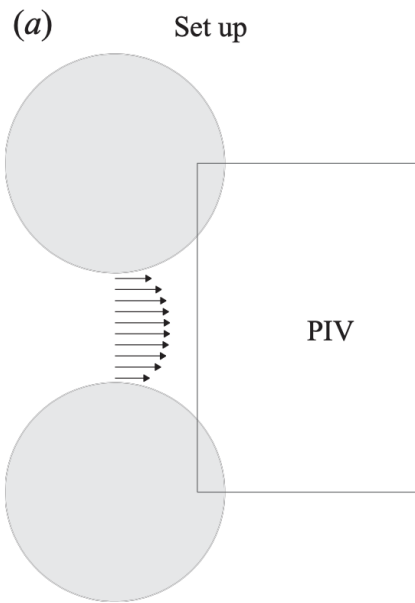
Obs.: The pseudo-inverse can **robustly** be computed from an **SVD** of $\mathbf{Q}^-$: **POD is still a fundamental step for DMD**

Dynamic Mode Decomposition

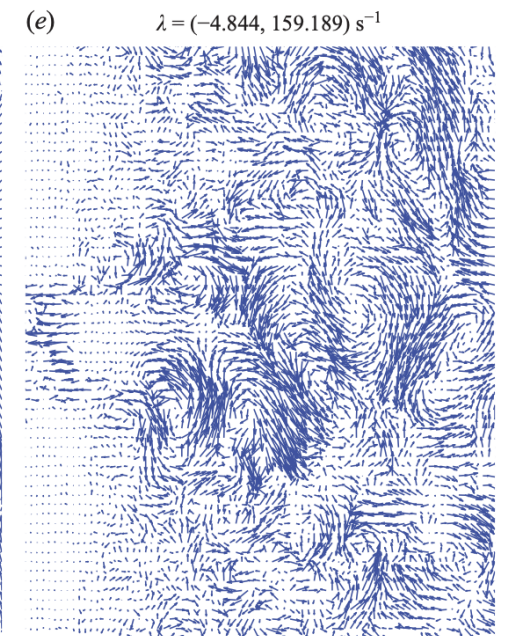
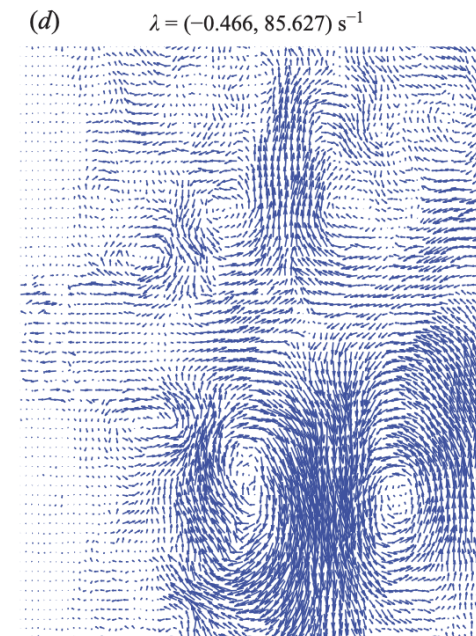
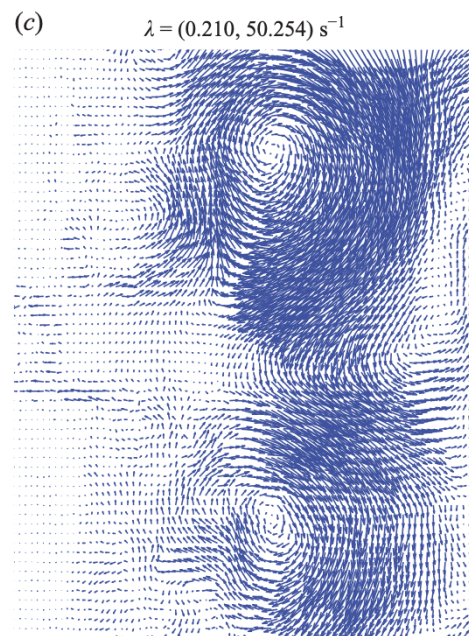
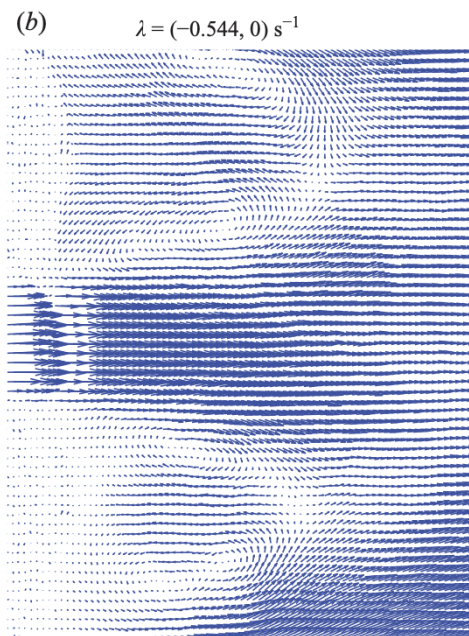
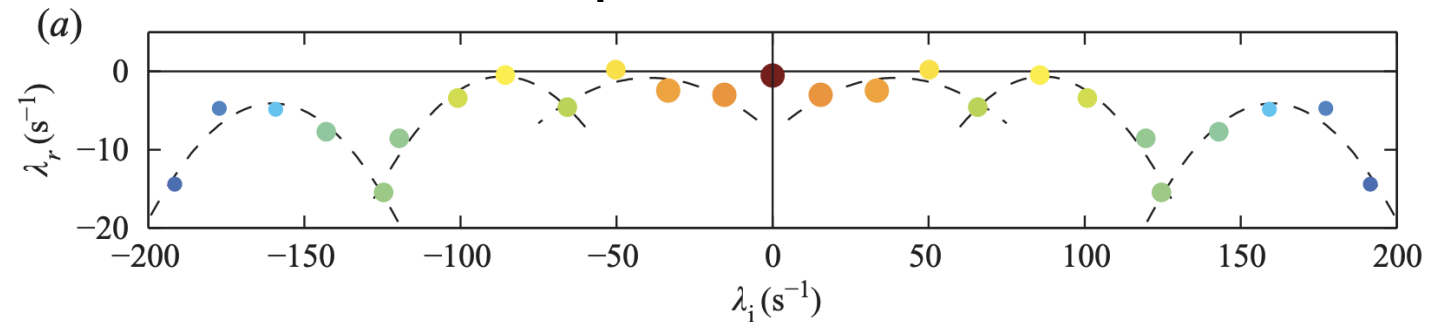


Brunton et al 2021

DMD: Example: PIV data from a jet



Spectrum of L



Schmid 2010

Linear System Identification (LSI)

As stated before, DMD (and also POD/SVD) have extensively been used in the several contexts, in particular for **Linear System Identification** (LSI), already a mature field around 1980-1990

Indeed, if we consider an autonomous linear system :

$$\begin{cases} \mathbf{x}_{n+1} = \mathbf{L}\mathbf{x}_n + \mathbf{B}\mathbf{u}_n & \in \mathbf{R}^N \\ \mathbf{y}_n = \mathbf{C}\mathbf{x}_n & \in \mathbf{R}^O \end{cases}$$

Linear System Identification (LSI)

Its solution, for an impulse excitation $\mathbf{u}_n = \hat{\mathbf{u}}\delta_{n,0}$ is:

$$\mathbf{x}_n = \mathbf{L}^{n-1}\mathbf{B}\hat{\mathbf{u}} \Rightarrow \mathbf{y}_n = \mathbf{C}\mathbf{L}^{n-1}\mathbf{B}\hat{\mathbf{u}}$$

LSI: How can we reconstruct $\mathbf{C}$ and $\mathbf{L}$ from the time-series $\mathbf{y}_n$?

The answer is given by the **Eigensystem Realization Algorithm** (**ERA**, Ho & Kalman 1965, Juang & Pappa 1985)

Linear System Identification (LSI)

We can arrange $\mathbf{y}_n$ in a Henkel-matrix structure (van Overschee 1996):

$$\mathbf{H} = \begin{bmatrix} \mathbf{y}_0 & \mathbf{y}_1 & \cdots \\ \mathbf{y}_1 & \mathbf{y}_2 & \cdots \\ \vdots & \vdots & \end{bmatrix} = \begin{bmatrix} \mathbf{C} \\ \mathbf{CL} \\ \mathbf{CL}^2 \\ \vdots \end{bmatrix} [\mathbf{B} \quad \mathbf{LB} \quad \mathbf{L}^2\mathbf{B} \cdots] \hat{\mathbf{u}}$$

If we perform the SVD (snapshot POD), and truncate it at “ r ”:

$$\mathbf{H} \approx \left(\mathbf{U}_r \mathbf{\Sigma}_r^{\frac{1}{2}} \right) \left(\mathbf{V}_r \mathbf{\Sigma}_r^{\frac{1}{2}} \right)^{*,T} = \tilde{\mathbf{U}}_r \tilde{\mathbf{V}}_r^{*,T} \approx \begin{bmatrix} \tilde{u}_{1,1}, \cdots, \tilde{u}_{1,r} \\ \vdots \\ \tilde{u}_{N,1}, \cdots, \tilde{u}_{N,r} \end{bmatrix} \begin{bmatrix} \tilde{v}_{1,1} & \cdots & \tilde{v}_{1,N} \\ \vdots & & \\ \tilde{v}_{r,1} & \cdots & \tilde{v}_{r,N} \end{bmatrix}$$

Linear System Identification (LSI)

By comparison :

- We can infer the matrix $\mathbf{C}$ to be the r -first rows of $\tilde{\mathbf{U}}$

$$\mathbf{C} = [\tilde{\mathbf{U}}]_{(1,N_o) \times (1,r)}$$

- Also, if we define:

$$\tilde{\mathbf{U}}^- = [\tilde{\mathbf{U}}]_{(N_o, \dots) \times (1,r)}, \quad \tilde{\mathbf{U}}^+ = [\tilde{\mathbf{U}}]_{(N_o+1, \dots) \times (1,r)}$$

We determine the matrix $\mathbf{L}$ to be $\mathbf{L} = \tilde{\mathbf{U}}^+ (\tilde{\mathbf{U}}^-)^\dagger$

We recover the DMD!!!

Obs.: We can only reconstruct the matrices $\mathbf{C}$ and $\mathbf{L}$ in the **observable space**, of dimension r !!

LSI Example: Wind Turbine Blade

$$MAC(\psi_1, \psi_2) = \frac{|\psi_1^H \psi_2|^2}{\psi_1^H \psi_1 \psi_2^H \psi_2}$$

Cadoret et al 2023

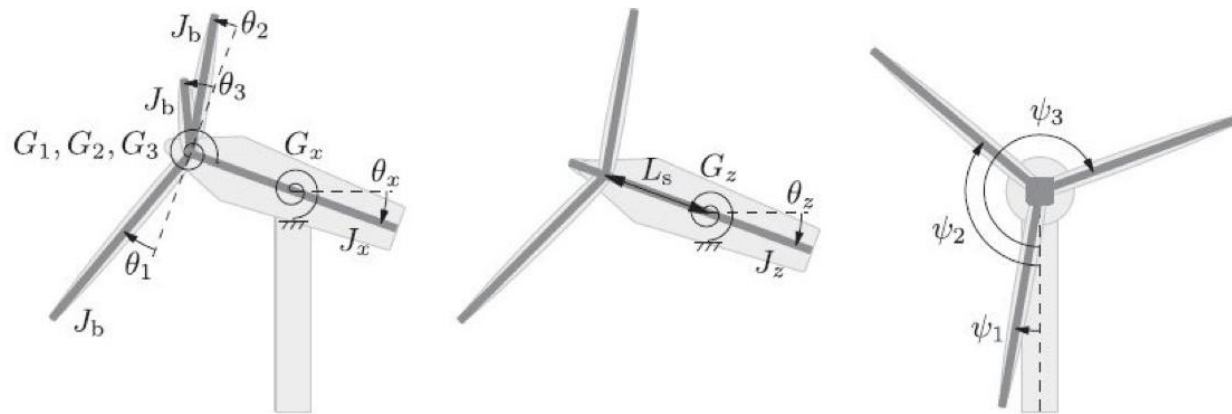


Fig. 1. Wind turbine model (Skjoldan, 2009)

Freq. (Hz)	σ_f (Hz)	Damp. (%)	σ_d (%)	MAC
1.698	0.003	0.833	0.173	0.999
1.479	0.003	1.160	0.188	0.994
1.256	0.003	1.295	0.260	0.999
1.825	0.003	0.627	0.142	0.998
1.600	0.003	0.498	0.096	0.999
1.378	0.002	0.585	0.124	0.998
0.643	0.001	0.714	0.179	0.989
0.867	0.002	0.431	0.156	0.986
1.092	0.007	0.578	0.630	0.987
0.747	0.001	0.329	0.098	1.000
0.671	0.001	0.270	0.093	1.000
0.453	0.005	1.399	0.734	0.994

Goal: to study the spectrum of L in time to **Monitor Structural Health**

Nonlinear Systems

Question: Is DMD truly justified for non-linear systems?

Since DMD is based on the idea of finding **linear** modes, one has to find ways to **linearize** dynamical systems, generally given as:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$$

Or, alternatively via its underlying flow, as:

$$\mathbf{x}_{n+1} = \mathbf{F}(\mathbf{x}_n) = \mathbf{x}_n + \int_{t_n}^{t_{n+1}} \mathbf{f}(\mathbf{x}(\tau)) d\tau$$

One way to do so is by using the **Koopman** operator theory (Rowley, Mezic et al 2011)

Nonlinear Systems: Koopman theory

Let $\phi(\boldsymbol{x})$ be an **observable**:

$$\phi: \boldsymbol{R}^N \rightarrow \boldsymbol{R} \text{ or } \boldsymbol{C}$$

Ex.: $\phi_1(\boldsymbol{x}) = \sin|\boldsymbol{x}|$, $\phi_2(\boldsymbol{x}) = x_1 + \tanh x_2^2$, $\phi_3(\boldsymbol{x}) = e^{i \cdot \text{sign}(x_1)}$, ...

The Koopman operator acts on such **observables** and is defined as:

$$\mathcal{K}\phi(\boldsymbol{x}) = \phi(\boldsymbol{F}(\boldsymbol{x}))$$

In other words, it is responsible for evaluating ϕ at a point in the state-space which is the **image** of $\boldsymbol{F}$ (the flow), starting from an initial condition $\boldsymbol{x}$

Nonlinear Systems: Koopman theory

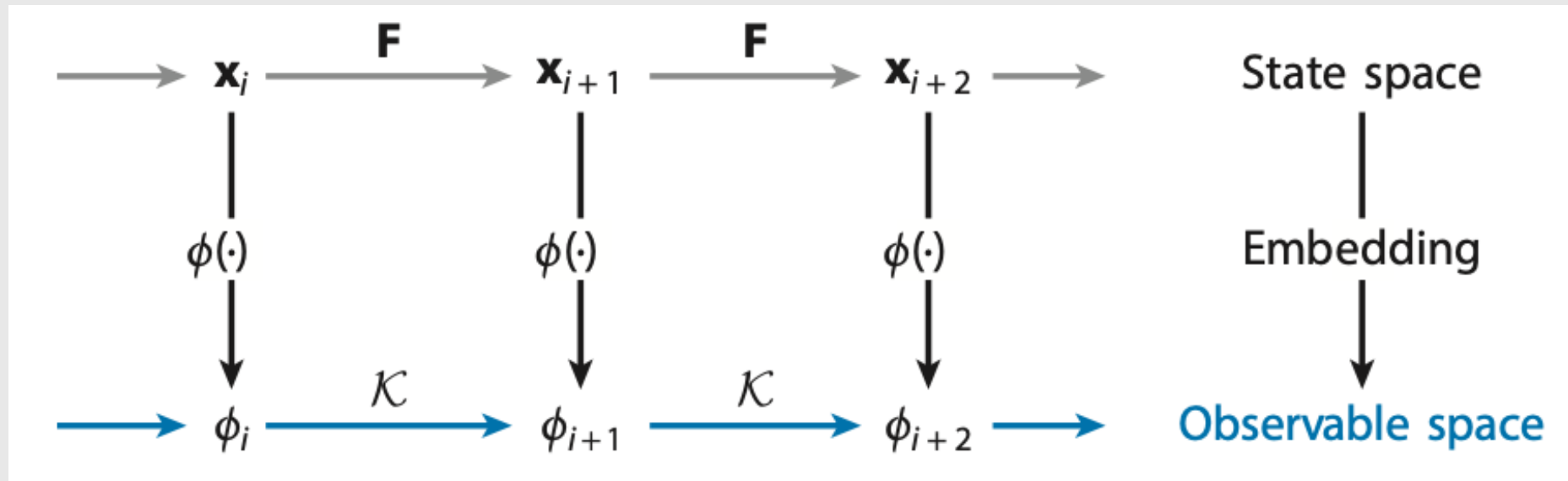
What is this good for?

The Koopman operator is **linear**:

$$\begin{aligned}\mathcal{K}(\alpha_1\phi_1(\mathbf{x}) + \alpha_2\phi_2(\mathbf{x})) &= \left(\alpha_1\phi_1(\mathbf{F}(\mathbf{x})) + \alpha_2\phi_2(\mathbf{F}(\mathbf{x}))\right) \\ &= \alpha_1\mathcal{K}\phi_1(\mathbf{x}) + \alpha_2\mathcal{K}\phi_2(\mathbf{x})\end{aligned}$$

Nonlinear Systems: Koopman theory

We have transformed a **non-linear** dynamics on a **finite-dimensional** space to a **linear** dynamics on a (possibly) **infinite-dimensional** space of observables



Schmid 2022

Nonlinear Systems: Koopman theory

Ex. 1: Let us consider the “**nonlinear**” system (Brunton et al 2016):

$$\dot{x}_1 = \mu x_1, \quad \dot{x}_2 = \lambda(x_2 - x_1^2)$$

Can we define a set of observables $\{y_j = y_j(\mathbf{x})\}$ that linearize the dynamics?

$$y_1 = y_1(\mathbf{x}) = x_1, \quad y_2 = y_2(\mathbf{x}) = x_2, \quad y_3 = y_3(\mathbf{x}) = x_1^2$$

The dynamics for y_j is **linear** and given by:

$$\dot{y}_1 = \mu y_1, \quad \dot{y}_2 = \lambda(y_2 - y_3), \quad \dot{y}_3 = 2x_1\dot{x}_1 = 2\mu y_3$$

Nonlinear Systems: Koopman theory

Ex. 2: Let us consider the **nonlinear** viscous Burgers equation:

$$\partial_t u + u \partial_x u = \nu \partial_{xx} u$$

The Cole-Hopf transformation:

$$y = y(u) = e^{u/2\nu}$$

Produces the following **linear** Heat equation:

$$\partial_t y = \nu \partial_{xx} y$$

Obs.: This is not always possible, for a given system

We should establish a **data-driven** methodology for it (**DMD**)

Nonlinear Systems: Koopman and DMD

From experiments or numerical simulations, we have access to several **vector-observables**:

$$\mathbf{d}_k = \mathbf{d}\left(F^k(\mathbf{x})\right)$$

That can be arranged in the data-matrices:

$$\mathbf{D}^- = [\mathbf{d}(\mathbf{x}), \dots, \mathbf{d}(F^{N_t-1}(\mathbf{x}))], \mathbf{D}^+ = [\mathbf{d}(F(\mathbf{x})), \dots, \mathbf{d}(F^{N_t}(\mathbf{x}))]$$

So that the relation between each other is given by:

$$\mathcal{K} \mathbf{D}^- = \mathbf{D}^+$$

Nonlinear Systems: Koopman and DMD

Obs.: The action of the Koopman operator is not like a matrix one!!

By computing

$$\mathbf{L} = \mathbf{D}^+ (\mathbf{D}^-)^\dagger = \boldsymbol{\phi}^T(\mathbf{x}) \boldsymbol{\Lambda} \boldsymbol{\phi}(\mathbf{x})$$

Where $\boldsymbol{\phi}(\mathbf{x}) = (\phi_1(\mathbf{x}), \dots)$ is a vector composed by Koopman eigen-functions:

$$\mathcal{K} \phi_j(\mathbf{x}) = \lambda_j \phi_j(\mathbf{x})$$

we determine only the projection of $\mathcal{K}$ (its eigen-functions) onto the observables dictionary

$$\mathbf{d}_k = \mathbf{d} \left(\mathbf{F}^k(\mathbf{x}) \right)$$

Nonlinear Systems: Koopman and DMD

Obs.: If the data dictionary data is nothing but the state space:

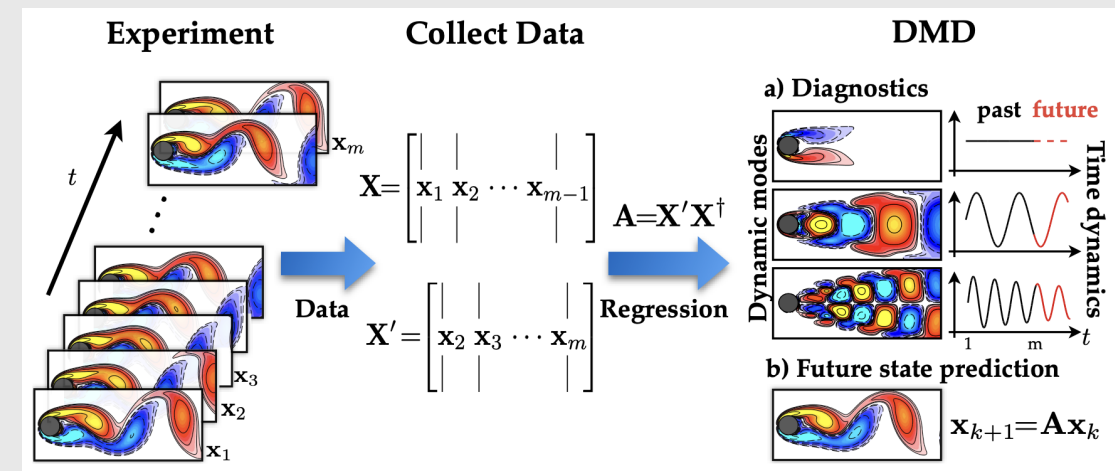
$$d(\mathbf{x}) = \mathbf{x}$$

Then,

$$\mathbf{D}^- = [\mathbf{x}, \dots, \mathbf{F}^{N_t-1}(\mathbf{x})]$$

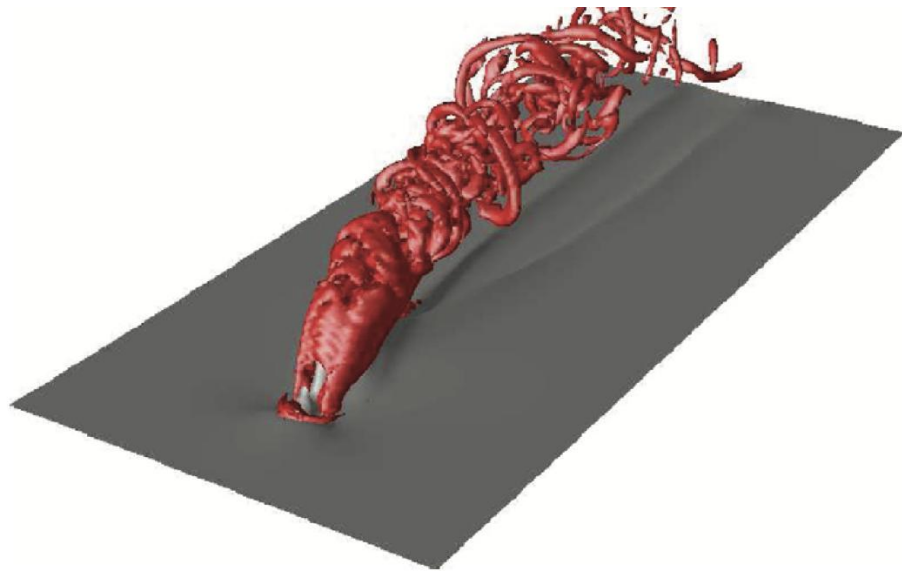
Is the classical snapshot-matrix

However, the data dictionary contains other nonlinear function of the state, the DMD is called **Extended DMD (EDMD, Williams et al 2015)**

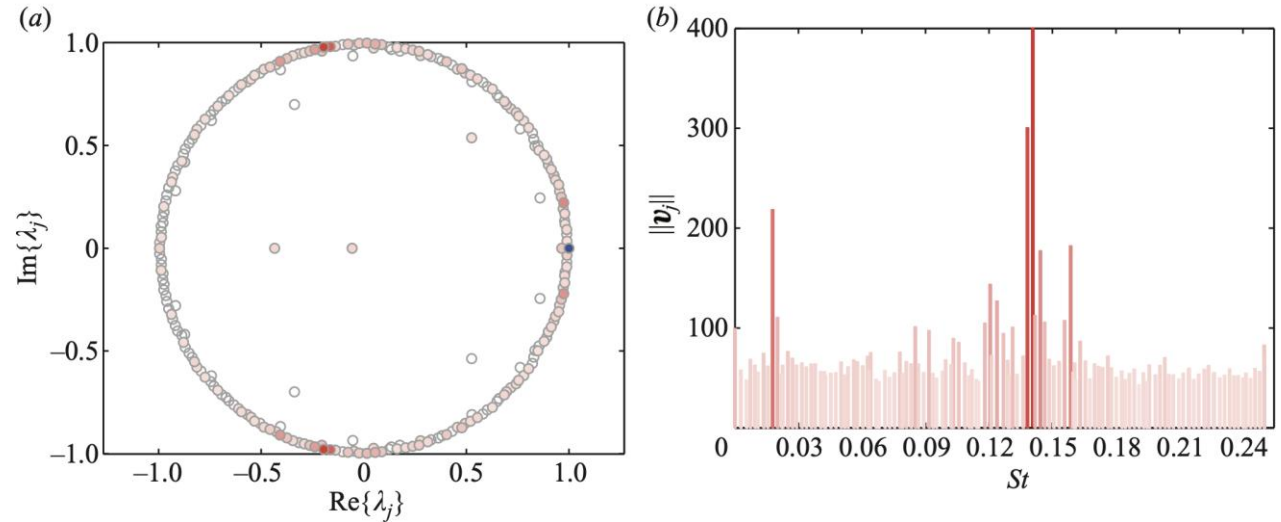


Brunton et al 2021

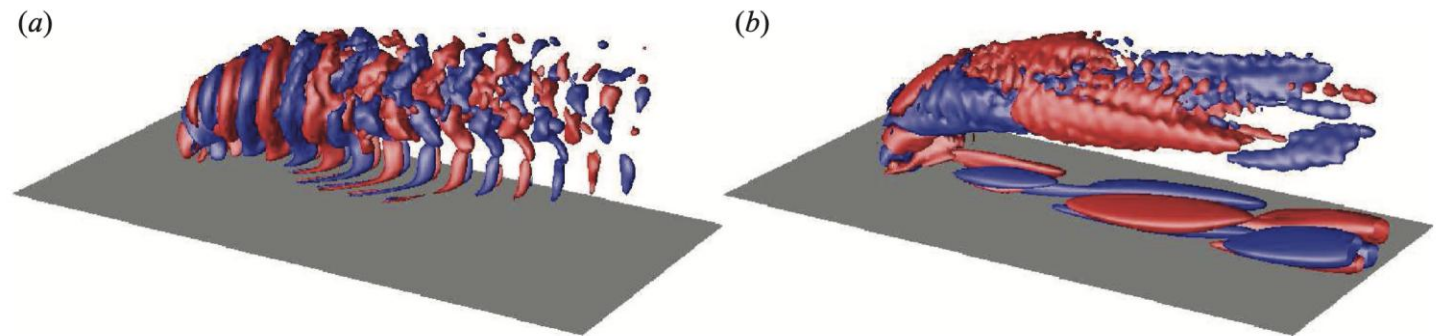
Koopman/DMD: Example Jet in crossflow



Rowley et al 2011



Spectrum of L



Some eigenvectors of L, Ψ

DMD with external forcing/control: DMDc

Let us consider that our system is given by:

$$\mathbf{q}(t_{k+1}) = \mathbf{L}\mathbf{q}(t_k) + \mathbf{B}\mathbf{u}(t_k)$$

As before (Proctor et al 2016), we want to find $\mathbf{L}$, and now $\mathbf{B}$, from data:

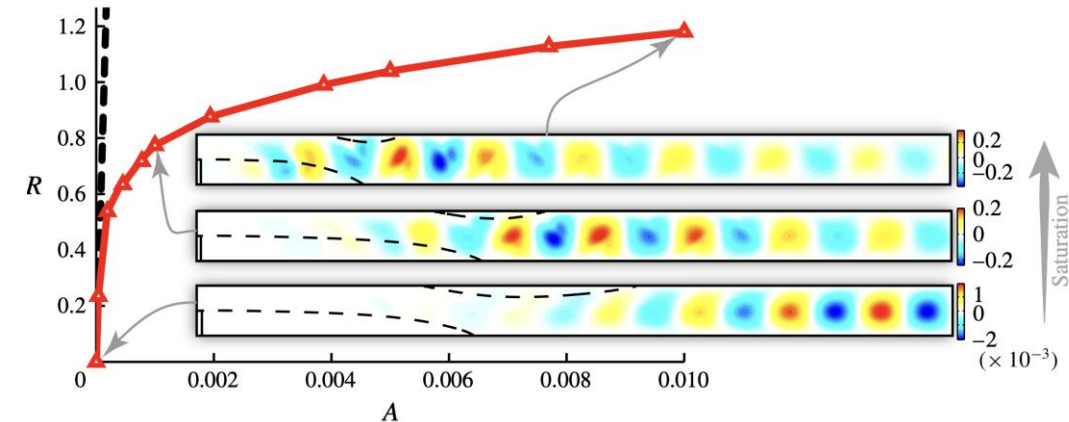
$$\mathbf{Q}^- = [\mathbf{q}(t_0), \dots, \mathbf{q}(t_{N-1})]$$

$$\mathbf{Q}^+ = [\mathbf{q}(t_1), \dots, \mathbf{q}(t_N)]$$

And

$$\mathbf{U} = [\mathbf{u}(t_0), \dots, \mathbf{u}(t_{N-1})]$$

This idea is very closely related to **ERA** for Linear Systems



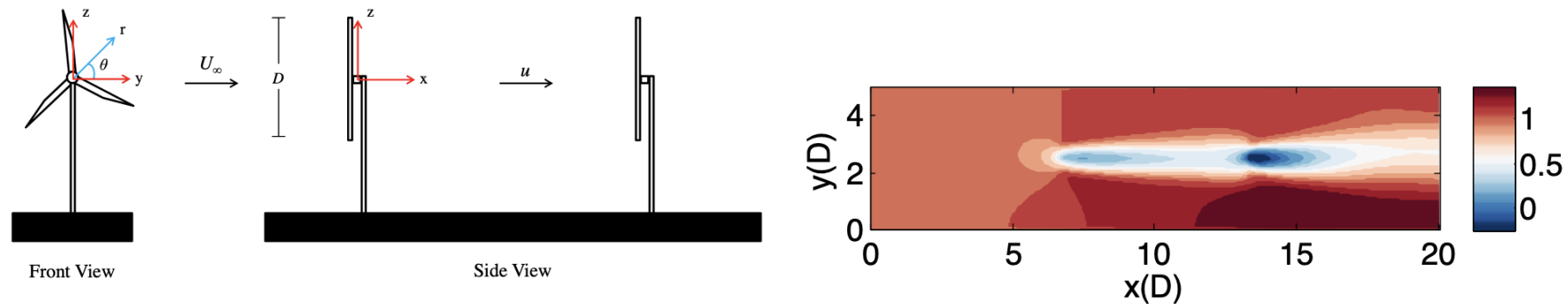
Mantic-Lugo, Gallaire 2016

Saturation process in a
Backward-Facing Step to a
harmonic forcing

DMDc: Application to Wind Farm control

Anoni et al, 2016, Cassamo & van Wingerden 2020

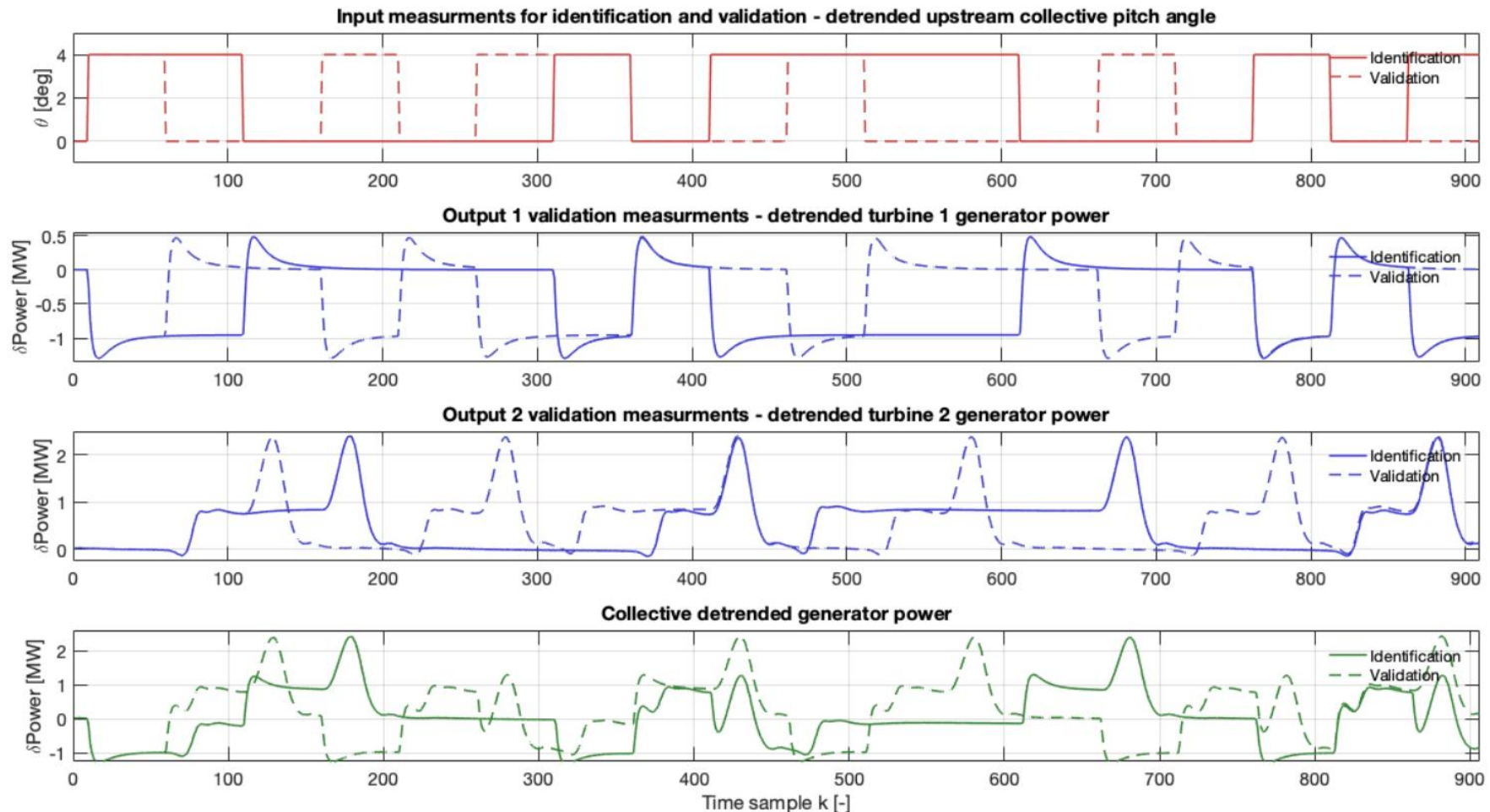
Two wind turbines



- Control $\mathbf{u}(t_k)$: turbine-related active variables: turbine global induction factor, blade pitch angles, etc

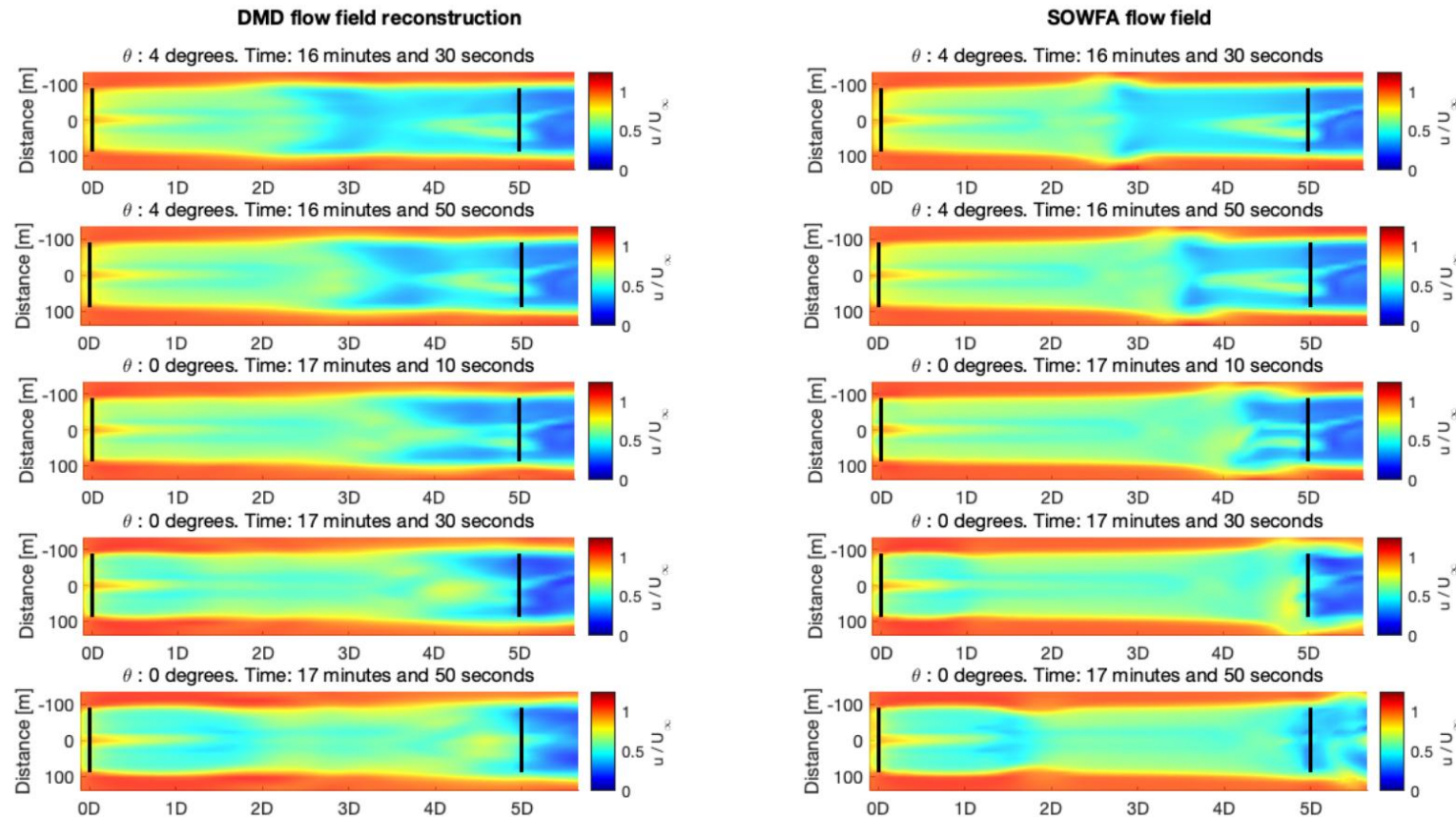
DMDc: Application to Wind Farm control

First Step: Data-acquisition: CFD



DMDc: Application to Wind Farm control

Second Step: Model learning using DMDc



Using Takens' embedding: Henkel-DMD

Oftentimes the whole state vector $\mathbf{x}$ is **not available**

If only $y = h(\mathbf{x}) \in \mathbf{R}$ is available, we may use a time-delay embedding (Takens 1980, Letellier & Aguirre 2002) :

$$\Phi(\mathbf{x}) = \left(h(\mathbf{x}), h(\mathbf{F}(\mathbf{x})), \dots, h(\mathbf{F}^{d-1}(\mathbf{x})) \right)^T = (y_0, y_1, \dots, y_{d-1})^T = \mathbf{y}^T$$

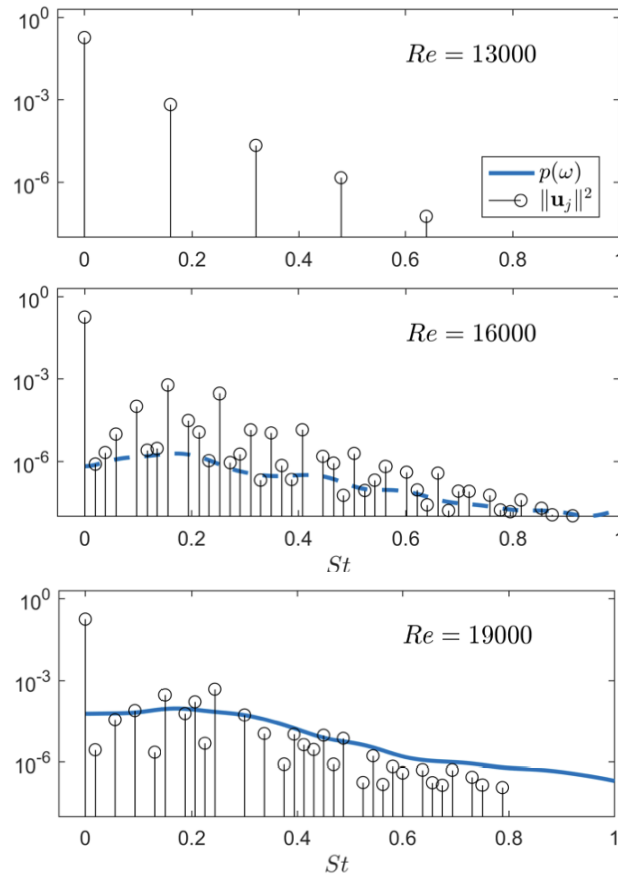
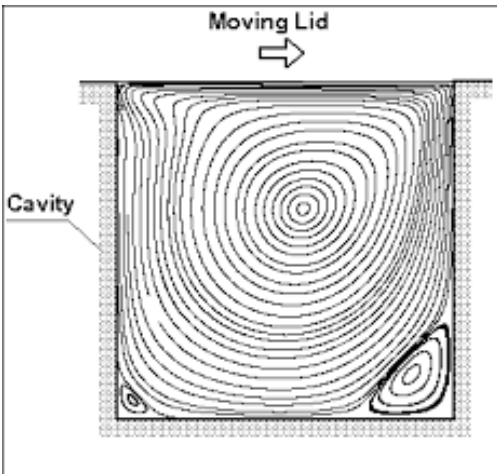
Takens: this is a diffeomorphism for sufficiently high d ($d > 2N$)

The “**snapshot**” matrix becomes a **Henkel** matrix (Arbabi Mezic 2017a):

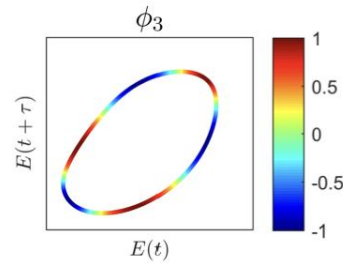
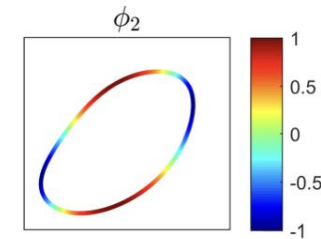
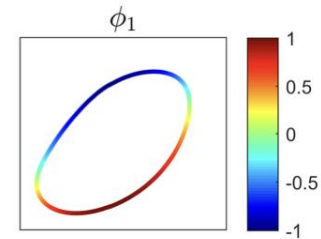
$$\mathbf{D} = [\mathbf{y}_0, \dots, \mathbf{y}_{N_t}] = \begin{bmatrix} y_0 & y_1 & \cdots \\ y_1 & y_2 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

Henkel-DMD : Reynolds-bifurcation in lid-driven cavity flow

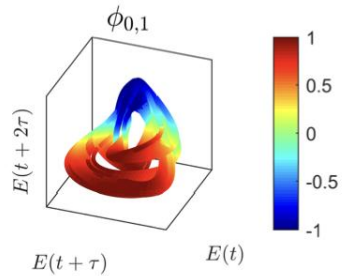
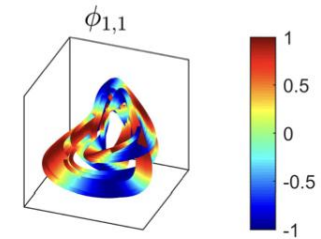
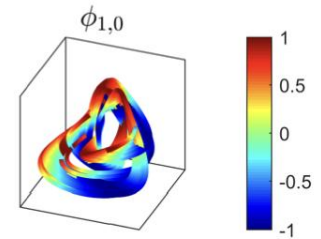
Arbabi Mezić 2017b



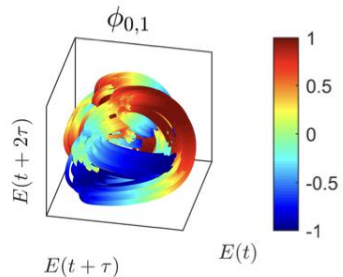
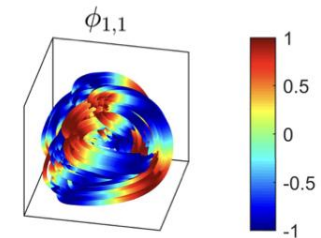
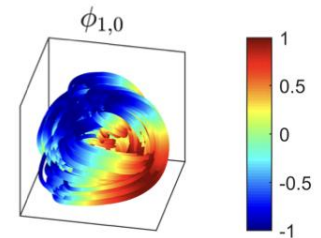
$Re = 13000$



$Re = 16000$

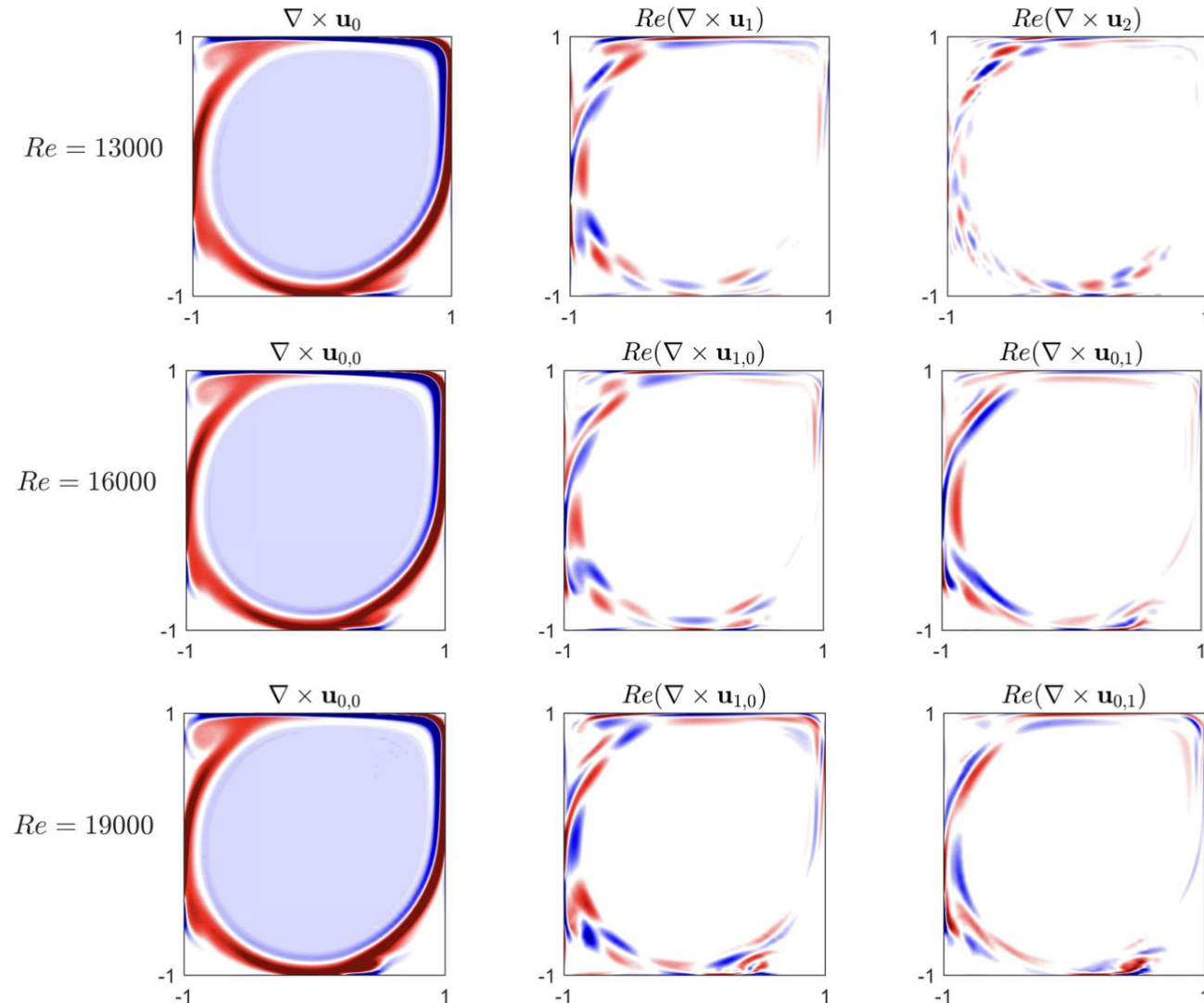


$Re = 19000$



Henkel-DMD : Reynolds-bifurcation in lid-driven cavity flow

Arbabi Mezić 2017b

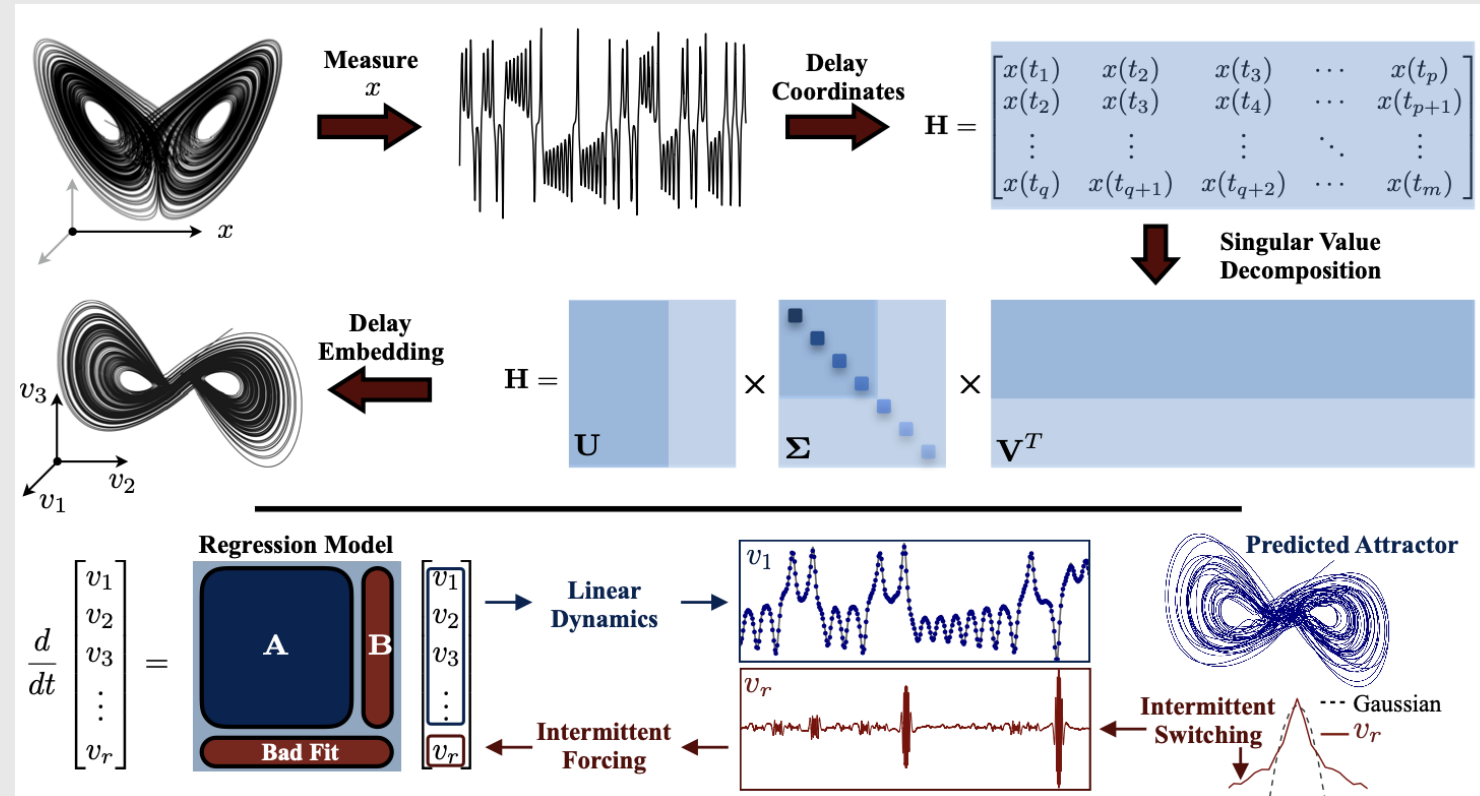


Henkel-DMD: Lorenz system

Brunton et al 2016:
Lorenz system

A linear model, made
from POD+DMD **does**
not capture **several**
fixed-points behavior
(wings of the butterfly)

An **intermittent**
external forcing term
should be added



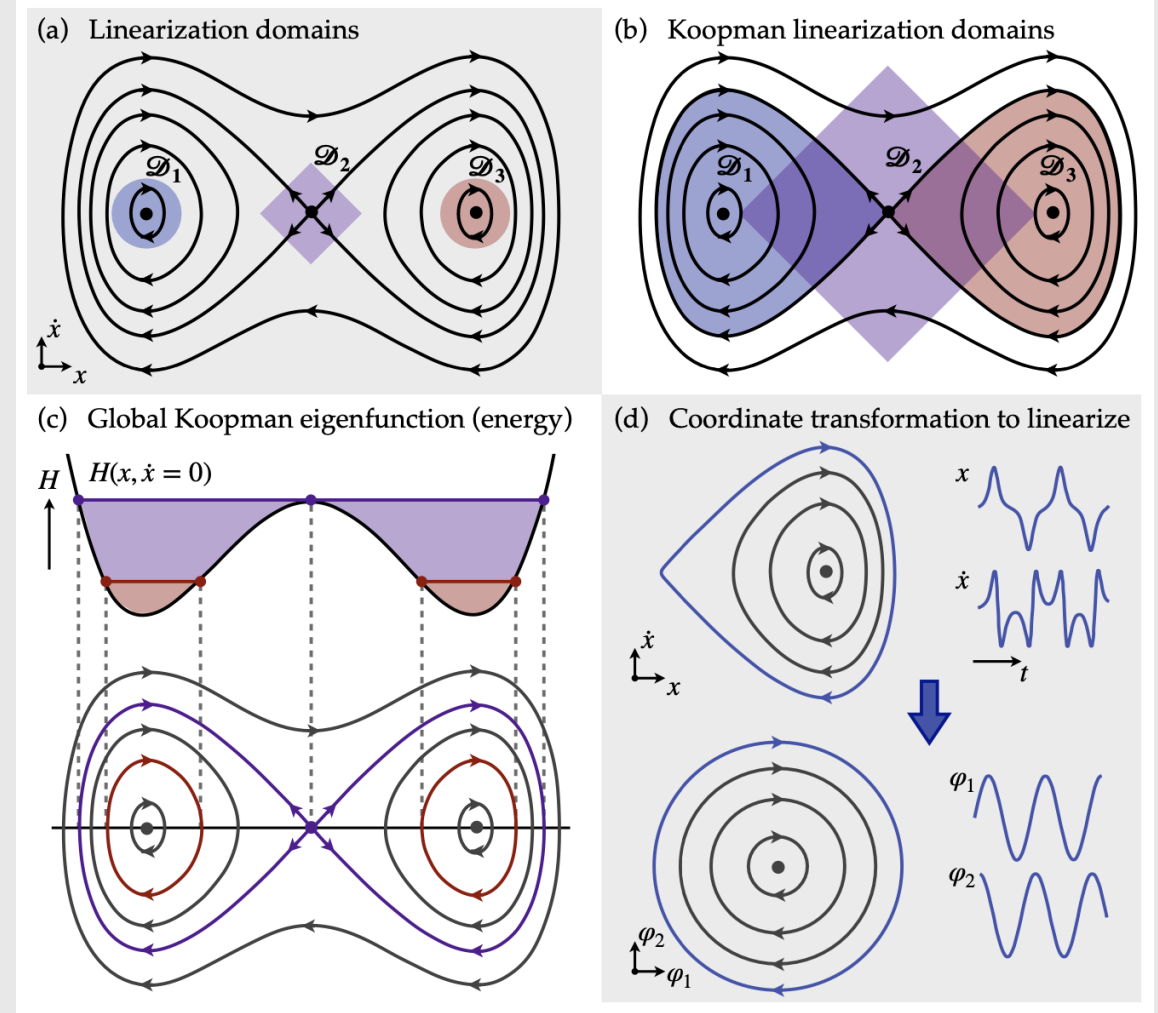
Limitations of DMD/Koopman

- Koopman operator provides a **more powerful linearization** than usual
- However, it is difficult to extend a KEF to the **entire state-space**
- It can still only provide a **local** description

Duffing Oscillator

$$\ddot{x} = x - x^3$$

Brunton et al 2021



Nonlinear System Identification

Instead of trying to identify **modes** and their **frequencies** from data $\mathbf{x}(t)$, let us directly identify $\mathbf{f}(\mathbf{x})$ such that:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$$

This is not only interesting when the dynamics is **unknown** (ex economics, biological systems, etc) but also when the model is **too complex** to handle easily (Fluid Mechanics, CFD, etc)

A vast literature on this topic is at hand (Aguirre Letellier 2009), including **Nonlinear AR/ARMA**

$$\mathbf{y}_{n+1} = \mathbf{g}(\mathbf{y}_n, \mathbf{y}_{n-1}, \dots)$$

NSI: Sparse Identification of Nonlinear Dynamical systems (SINDy)

A popular tool has recently gained attention (Sparse Identification of Nonlinear Dynamical systems- SINDy, Brunton et al 2016, Champion et al 2019)

This relies on the philosophical principle that $f(\mathbf{x})$ should be as “**simple**” as possible

If we give ourselves a dictionary of non-linear functions:

$$\Theta(\mathbf{x}) = [1 \quad \mathbf{x} \quad \mathbf{x}^{P_2} \quad \cdots \quad \sin \mathbf{x} \quad \cos \mathbf{x} \quad \cdots]$$

We find

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}(t)) \approx \Theta(\mathbf{x}(t))\xi$$

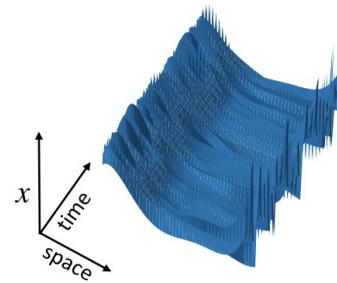
As to minimize a classical quadratic error functional **and** a **Sparsity-Promoting penalization**:

$$J(\xi) = \frac{1}{2} \|\dot{\mathbf{x}} - \Theta(\mathbf{x}(t))\xi\|^2 + \lambda |\xi|_1$$

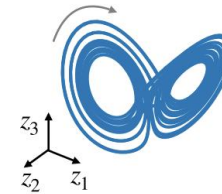
NSI: Sparse Identification of Nonlinear Dynamical systems (SINDy)

Brunton et al 2016
A few examples

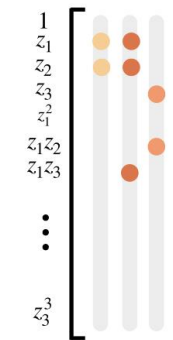
A Lorenz



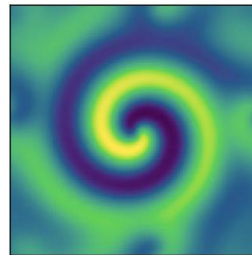
$$\begin{aligned}\dot{z}_1 &= -10.0z_1 + 10.0z_2 \\ \dot{z}_2 &= 27.7z_1 - 0.9z_2 - 5.5z_1z_3 \\ \dot{z}_3 &= -2.7z_3 + 5.5z_1z_2\end{aligned}$$



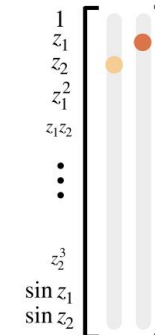
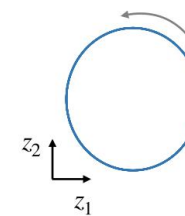
matrix



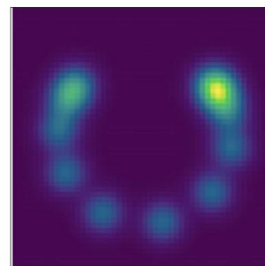
B Reaction-diffusion



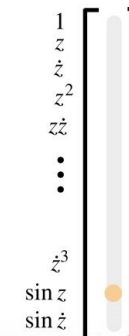
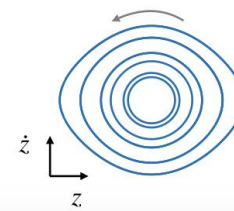
$$\begin{aligned}\dot{z}_1 &= -0.85z_2 \\ \dot{z}_2 &= 0.97z_1\end{aligned}$$



C Nonlinear pendulum



$$\dot{z} = -0.99 \sin z$$



Summary

- Data-Driven Dynamics and System Identification is both a **mature** but a **still evolving** field of research
- Although some of their techniques still lack theoretical justification, for **nonlinear systems**, it can be (and had been) applied in **many practical applications**

Perspectives for Offshore Mechanics

In Vortex Induced Vibration (VIV) phenomena:

$$\mathbf{x} = \begin{pmatrix} \mathbf{x}_f \\ \mathbf{x}_s \end{pmatrix} \Rightarrow \frac{d}{dt} \begin{pmatrix} \mathbf{x}_f \\ \mathbf{x}_s \end{pmatrix} = \mathbf{f} \begin{pmatrix} \mathbf{x}_f \\ \mathbf{x}_s \end{pmatrix}$$

- $\mathbf{x}_f$: fluid variables
- $\mathbf{x}_s$: solid variables

Obs.: Oftentimes experiments only provide solid-related quantity $y = h(\mathbf{x}_s)$, such as its displacement, velocity, etc

Q.: Can we reconstruct the dynamics $\mathbf{f}(\mathbf{x})$ projected onto

$$\Phi(\mathbf{x}) = \left(h(\mathbf{x}), h(\mathbf{F}(\mathbf{x})), \dots, h(\mathbf{F}^{d-1}(\mathbf{x})) \right)^T = (y_0, y_1, \dots, y_{d-1})^T = \mathbf{y}^T$$

Perspectives for Offshore Mechanics

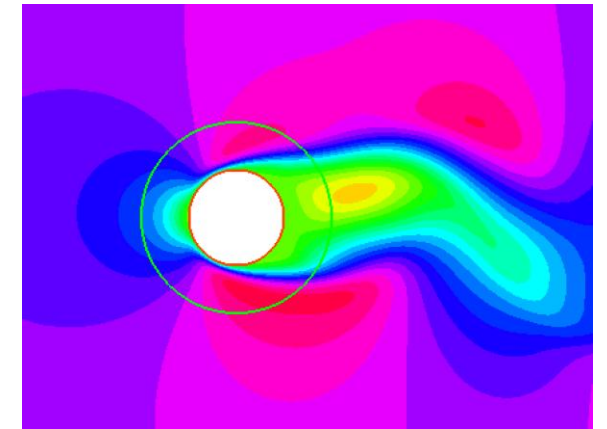
For example, let us consider the VIV around a rigid cylinder at:

- Reynolds number $Re = U_\infty D / \nu = 100$
- Reduced velocity $U_r = U_\infty f_n / D \in (4,9)$

Q.: Can we characterize the dynamics using solid-only information ?

Q.: Can we build a reliable dynamical model for it valid for several parameters (ex $U_r \in (4,9)$) ?

Q.: Can we bypass the need for CFD/experiments for describing x_s ?

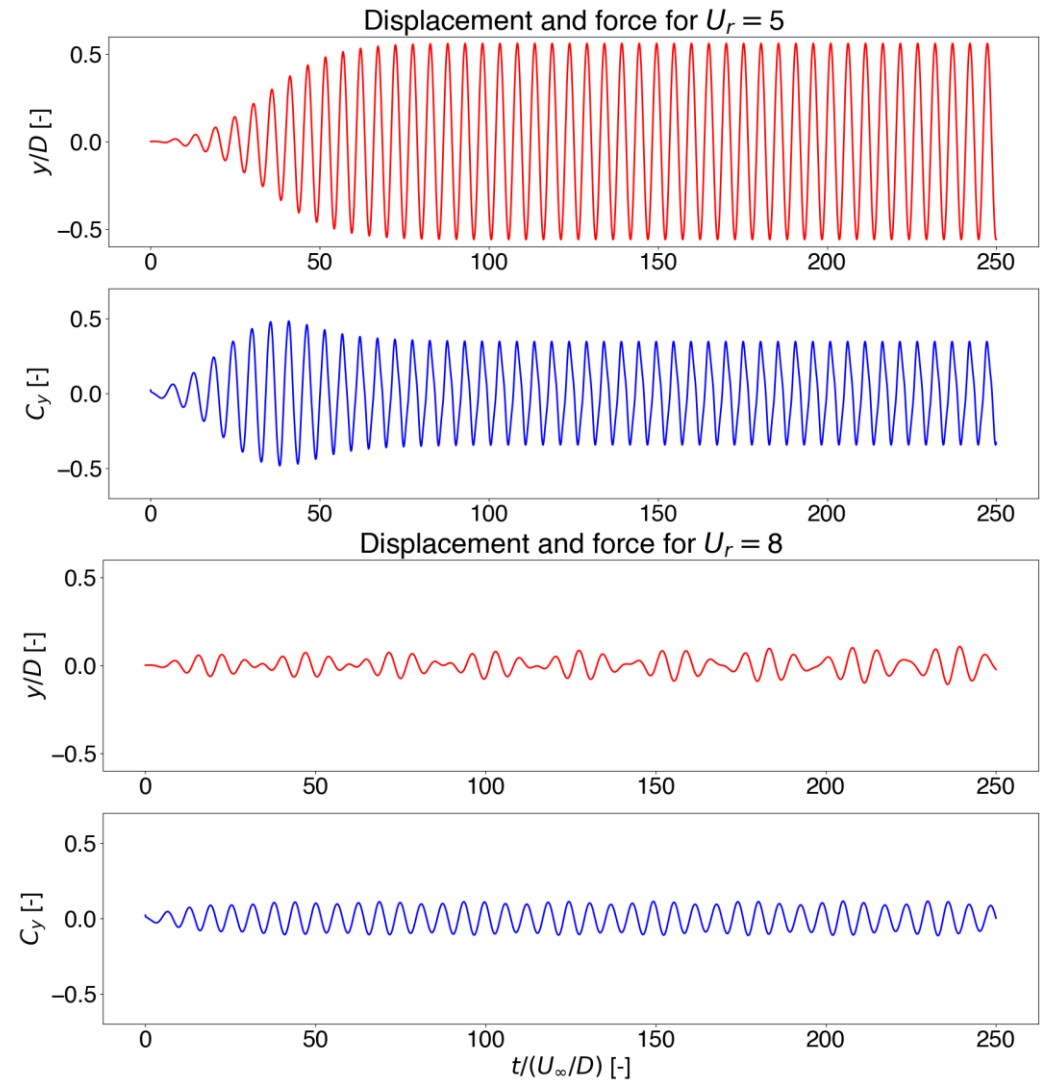
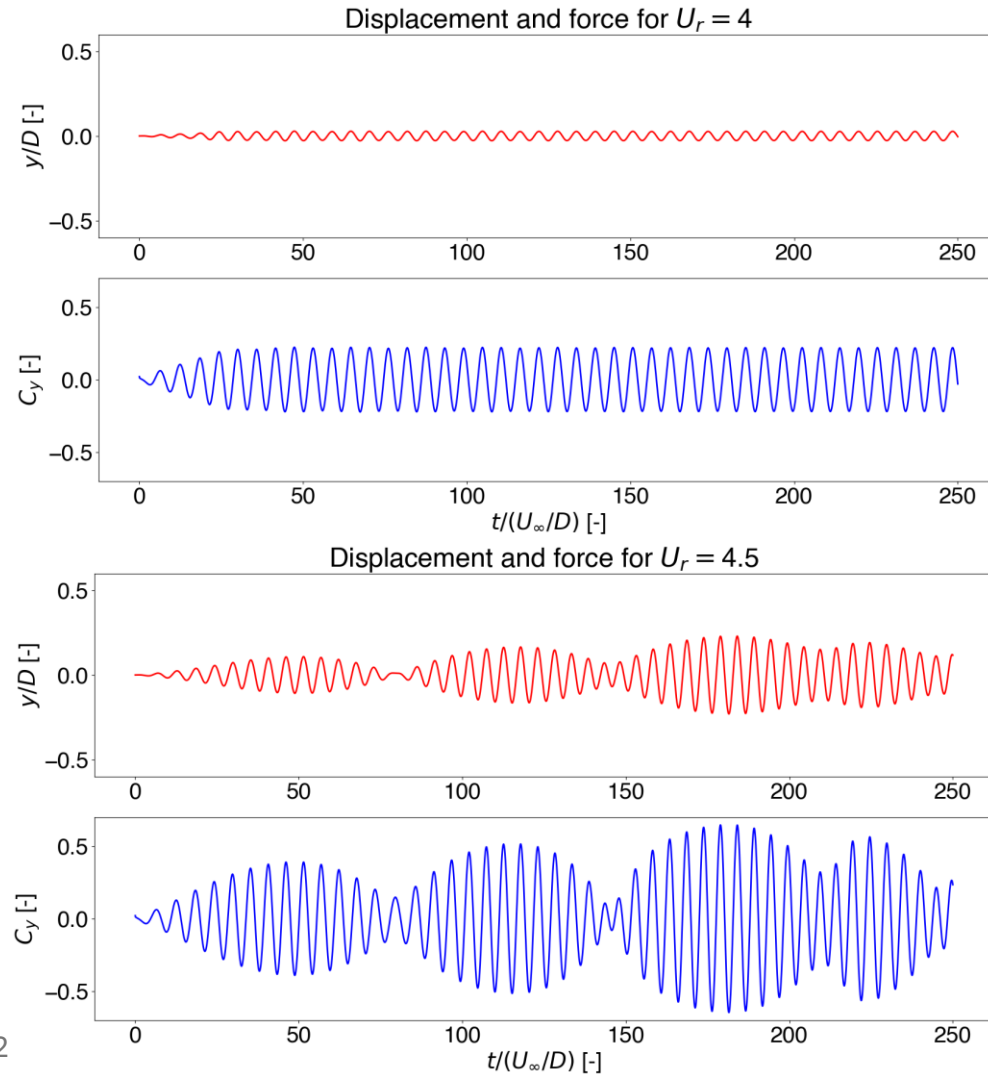


Flow visualization

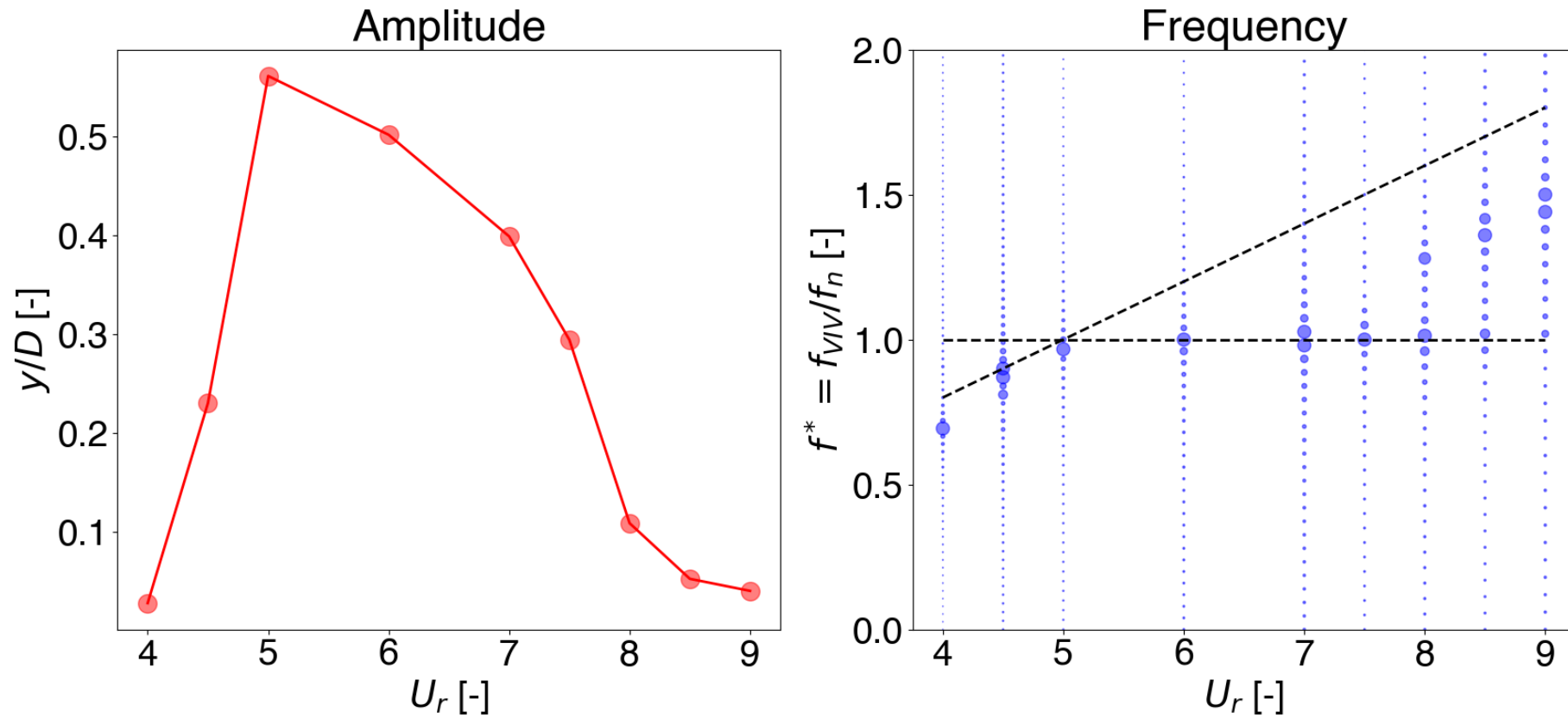
CFD results

FreeFEM++

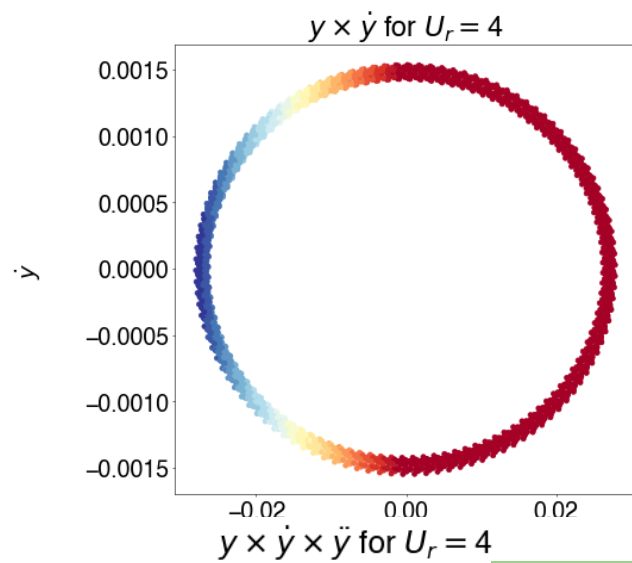
Perspectives for Offshore Mechanics



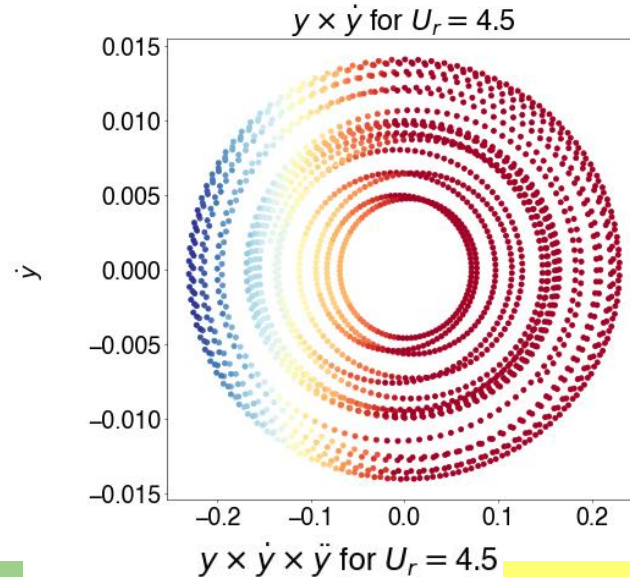
Perspectives for Offshore Mechanics



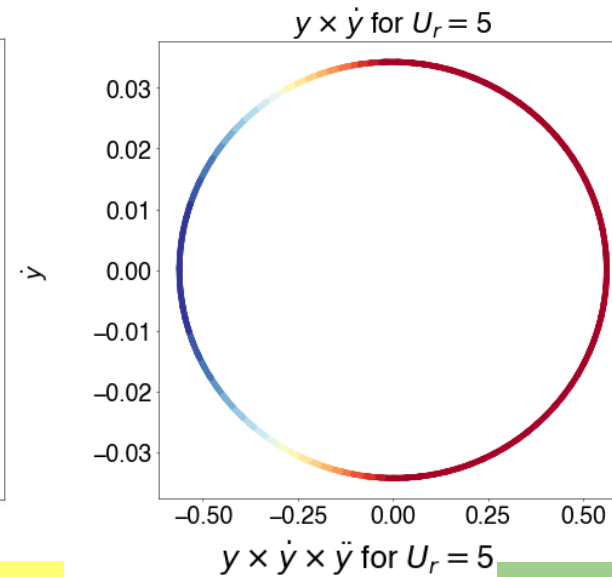
Perspectives for Offshore Mechanics



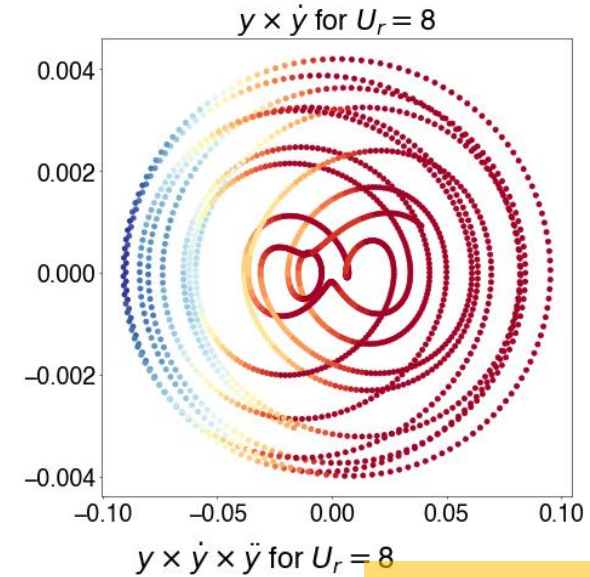
Easy!!



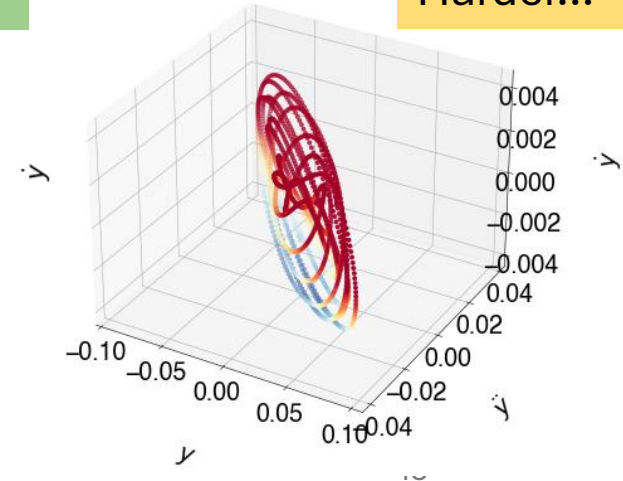
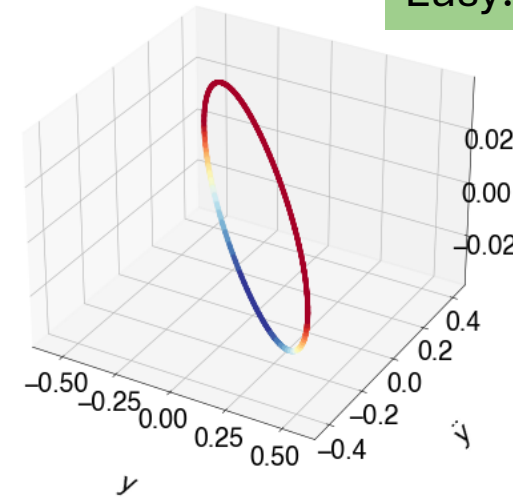
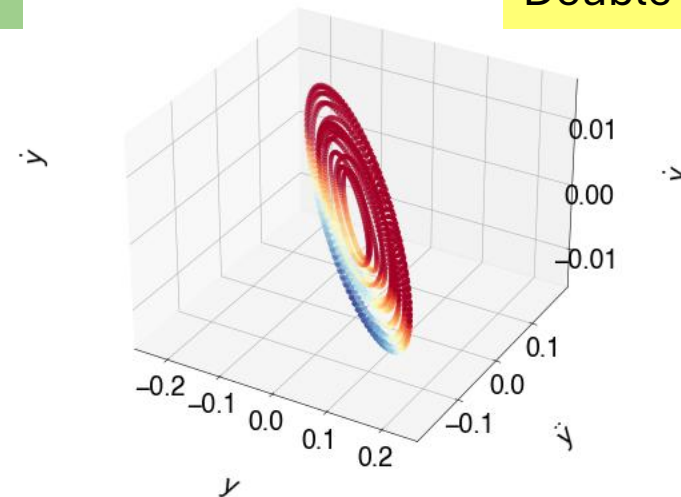
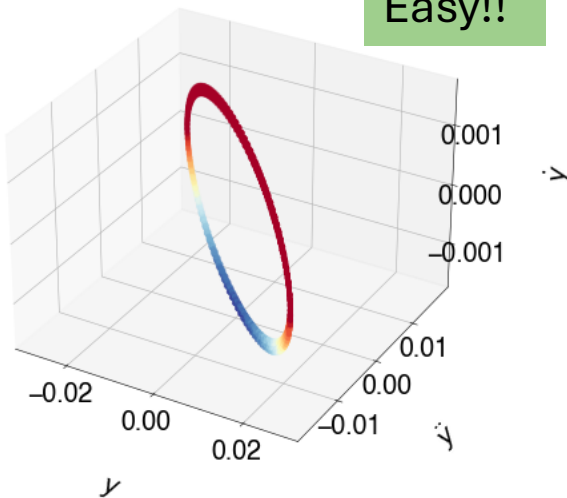
Doable



Easy!!



Harder...



Perspectives for Offshore Mechanics

Another example, let us consider the **imposed** movement of an object for (possibly time-varying) velocity

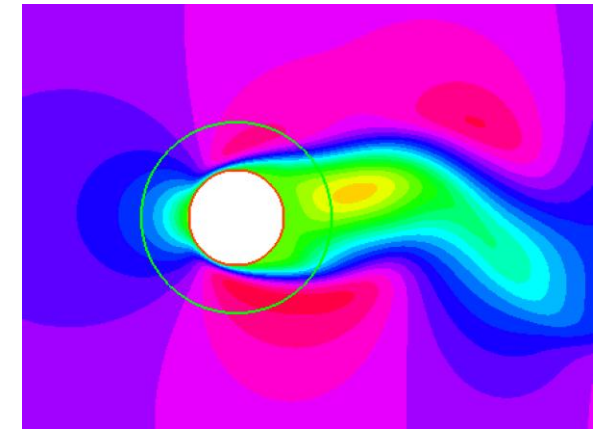
$$u(t) = U_{\infty}(t)$$

Q.: Can we characterize the dynamics of

$$\begin{aligned}\Phi(\mathbf{x}) &= \left(h(\mathbf{x}), h(\mathbf{F}(\mathbf{x})), \dots, h(\mathbf{F}^{d-1}(\mathbf{x})) \right)^T \\ &= (F_0, F_1, \dots, F_{d-1})^T\end{aligned}$$

using, for example, DMDc?

Q.: Can we build a dynamical system describing, **from data**, the dynamics of the force acting upon the object (Morrisson)?



Flow visualization

CFD results

FreeFEM++

Perspectives for Offshore Mechanics

More results are coming shortly....

Thank you for your attention